

Dynamic Macroeconomics II

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Fiscal Policy: Ricardian Equivalence Basic Model

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Basic model

- Consider an infinite horizon economy with no production. Each agent seeks to maximize her intertemporal utility:

$$\sum_{t=0}^{\infty} \beta^t u(c_t)$$

where $\beta \in (0, 1)$ and $u(\cdot)$ is a strictly increasing, strictly concave and twice differentiable utility function.

- Inada condition: $\lim_{c \rightarrow 0} u'(c) = +\infty$.
- There is no uncertainty in the economy.
- Households can smooth consumption between periods by investing in a single, risk-free asset that bears a fixed gross one-period rate of return $R > 1$.
- In each period t , consumers face the following sequential budget constraint:

$$c_t + \frac{b_{t+1}}{R} \leq y_t + b_t \quad (1)$$

- Let $\{y_t\}_{t=0}^{\infty}$ be a non-stochastic endowment sequence, $y_t \geq 0$, such that

$$\sum_{t=0}^{\infty} \beta^t y_t < \infty$$

- Agents choose the asset holding sequence $\{b_t\}_{t=1}^{\infty}$. b_0 is taken as given by the consumer.
- Assume $R\beta = 1$.



There are **two possible restrictions** that we may impose on the consumer:

- 1 If $b_{t+1} \geq 0$, households are allowed to lend but not to borrow. A negative b_{t+1} represents borrowing as it increases current consumption:

$$c_t \leq y_t + b_t - (1/R)b_{t+1}$$

- 2 An **alternative** is to **let consumers borrow an amount that is feasible to pay**. Impose $c_t = 0$ on sequential budget constraint $\forall t$. Solve for b_t in constraint for period t . Iterate forward to get:

$$\tilde{b}_t = - \sum_{j=0}^{\infty} R^{-j} y_{t+j} = -PV(y)$$

This level is called the **Natural Debt Limit (NDL)** (Aiyagari 1994). The alternative constraint is

$$b_{t+1} \geq \tilde{b}_{t+1} < 0$$

The operation we just did (excluding setting $c_t = 0$) is called "Solving Forward": to write b_t as a function of future variables. Steps:

- 1 Take sequential constraint in t , solve for b_t
- 2 Take constraint in t , iterate forward to get constraint in $t + 1$.
- 3 Solve for b_{t+1} from constraint in $t + 1$.
- 4 Substitute into seq. constraint in t .
- 5 Iterate: take b_{t+2} from constraint in $t + 2$, substitute into seq. constraint in t .
- 6 Repeat...
- 7 Assume

$$\lim_{T \rightarrow \infty} R^{-T} b_{t+T} = 0$$

This is the Transversality Condition.

Intuition for NDL:

- Debt cannot be larger than present value of future income.
- Even making consumption very small, consumer cannot repay debt larger than $\tilde{b}_{t+1}$

Intuition for Transversality Condition:

- Discounted value of “last” asset choice is zero.
- Eliminates Ponzi scheme: to borrow today amount equivalent to principal and interest of previous debt, never repay.

Also, note that:

- $b_{t+1} \geq \tilde{b}_{t+1}$ is a weaker restriction than $b_t \geq 0$.
- We assume there is no default risk.
- Assumption $R\beta = 1$ implies, in model with no borrowing limits, $c_t = c_{t+1} = \dots = \bar{c}$ (perfect consumption smoothing).

- Given b_0 and R , the sequences c_t and b_{t+1} solve the consumer's problem, in which we **impose no borrowing**:

$$\max_{\{c_t, b_{t+1}\}_{t=0}^{\infty}} \sum_{t=0}^{\infty} \beta^t u(c_t)$$

s.t.

$$\begin{aligned} c_t + \frac{b_{t+1}}{R} &\leq y_t + b_t & \forall t \\ b_{t+1} &\geq 0 & \forall t \end{aligned}$$

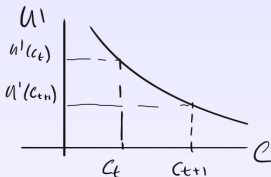
- Lagrangian of this problem:

$$\mathcal{L} = \sum_{t=0}^{\infty} \left[\beta^t u(c_t) + \lambda_t \left(y_t + b_t - c_t - \frac{b_{t+1}}{R} \right) + \mu_t b_{t+1} \right]$$

- Lagrange multiplier λ_t will **always** be strictly positive because utility is strictly increasing.

- From first order conditions, we obtain modified version of Euler equation:

$$u'(c_t) = \beta R u'(c_{t+1}) + \frac{R\mu_t}{\beta^t}$$



- In general $\mu_t \geq 0$: it is a Lagrange multiplier.
- If consumer wants to borrow, but we do not allow him, then $\mu_t > 0$
- I.e. the constraint becomes active. The consumer gets as close as possible to borrowing, but can only achieve $b_{t+1} = 0$.
- If $\mu_t > 0$, then $u'(c_t) > \beta R u'(c_{t+1})$.
- Then, as $R\beta = 1$, $c_t < c_{t+1}$, because of decreasing marginal utility.
- Seq. budget constraint tell us $c_t = y_t + b_t$

- **On the other hand**, if the consumer does not want to borrow, the constraint is not active.
- Then $\mu_t = 0$ and $u'(c_t) = \beta R u'(c_{t+1}) \implies c_t = c_{t+1} \quad \forall t$.

Note that:

- Perfect consumption smoothing in the absence of borrowing constraints is a consequence of the concavity of $u(\cdot)$.
- As we will see in further examples, consumption plan depends on the sequence of $\{y_t\}_{t=0}^{\infty}$.
- Whether the borrowing constraint binds or not **depends on the characteristics of the income path**.

Example 1

- Suppose that $b_0 = 0$ and that $y_t = \{y_h, y_\ell, y_h, y_\ell, \dots\}$, where $y_h > y_\ell$. Consumer faces constraint $b_{t+1} \geq 0$.
- What does consumer want to do in first period? Borrow or save?

- Consumer wants to save. She is not constrained.
- This will imply perfect smoothing. We calculate the **level** of consumption by solving

$$PV(y) = PV(c)$$

- I.e. we go from the set of sequential constraints to the unique intertemporal constraint, and solve for $\bar{c}$. **Check** you can do it, need to impose Transversality.
- Discounted value of endowment is (recall $R\beta = 1$) (**check**):

$$\sum_{t=0}^{\infty} \beta^t y_t = \sum_{t=0}^{\infty} \beta^{2t} (y_h + \beta y_\ell) = \frac{y_h + \beta y_\ell}{1 - \beta^2}$$

- Using the Euler equation and the sequential budget constraint, we obtain an expression for consumption:

$$\frac{\bar{c}}{1 - \beta} = \frac{y_h + \beta y_\ell}{1 - \beta^2} \implies \bar{c} = \frac{y_h + \beta y_\ell}{1 + \beta}$$

- $c_t = \bar{c} \quad \forall t \geq 0$

- Once we have calculated optimal consumption plan, we can construct associated savings plan.
- Using the budget constraint, we can show (**check**) that the optimal sequence of bond holdings is given by:

$$b_{t+1} = \begin{cases} (y_h - y_\ell)/(1 + \beta) > 0 & \text{for } t = 0, 2, 4, 6, \dots \\ 0 & \text{for } t = 1, 3, 5, 7, \dots \end{cases} \quad (2)$$

- The consumer is never constrained.
- We can calculate value of multipliers using Euler equation and optimal consumption sequence:

$$\mu_t = \beta^t R^{-1} u'(c_t) - \beta^{t+1} u'(c_{t+1})$$

Example 2

- Again, assume $b_0 = 0$, but sequence of endowments is given by $\{y_\ell, y_h, y_\ell, y_h, \dots\}$, where $y_h > y_\ell > 0$. Consumer faces constraint $b_{t+1} \geq 0$.
- The optimal path for consumption is $c_0 = y_\ell$ and from period 1 onward the optimal consumption stream is the same as in the previous example.
- For asset holdings, $b_1 = 0$ and b_{t+1} follows equation (2) $\forall t \geq 1$.
- The consumer faces borrowing constraint only in period 0.

Example 3

$$PV(y) < \infty \quad R > \lambda > 1$$

$$\sum R^{-t} y_t = \sum R^{-t} \lambda^t = \sum (\beta \lambda)^t$$

- Suppose $b_0 = 0$ and $y_t = \lambda^t \quad \forall t$, where $1 < \lambda < R$ (bad notation, we had used λ_t for the budget multiplier!).
- The last assumption implies that $\lambda\beta < 1$.
- Consumer faces constraint $b_{t+1} \geq 0$.
- $\lambda > 1$ means that the economy grows and $\lambda < R$ ensures that the discounted value of endowment stream converges.
- Consumer would like to bring future income to current date.
- Optimal plans for consumption and bonds are $c_t = y_t = \lambda^t$ and $b_{t+1} = 0 \quad \forall t$.
- Consumer is always borrowing constrained.